When the AI Label Is Wrong

Do explanations help people catch it, and how do they make people feel? — Design Summary (v3)

CEN5728 Advanced HCI, Group 4

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1. The Idea and the Gap

In Deepthi's proposal, participants read made-up posts from a fictional town. Each post carries a scripted AI label, "Likely accurate" or "Likely misleading", and some labels are deliberately wrong. The area is crowded, so the gap is the **combination**: false alarms and misses compared directly, four explanation depths including a counterfactual, and affect (tension, frustration) measured alongside trust and accuracy. The full literature search must confirm it.

Decision history: loan approval (proposal) → agent permission prompts (v2.1, archived) → content labels (v3). It needs Prof. Hamam's approval and ethics approval.

2. The Study

Each participant judges 24 posts in 4 sets of 6, one set per condition, with a fact sheet per post, clicking Agree or Disagree with the label. 8 labels are wrong (4 false alarms, 4 misses), with error type and difficulty fully crossed.

	Condition	Under the label
A	No explanation	The label only
B	Static reason	One fixed sentence arguing for the label
C	Interactive "Why?"	Source, Claim vs record (true ✓/✗), Wording rows
D	Counterfactual	"Would be labeled … if …" sentence

RQ1: Do explanations help people catch wrong labels, and does it differ for false alarms vs misses?

- **H1:** detection of wrong labels differs by condition.
- **H1a (headline):** misses are caught less often than false alarms.
- **H1b:** the gap between error types is smallest in C.

RQ2: Which explanation type gives the best error detection?

- **H2:** detection is highest in C and lowest in B; D is exploratory.

RQ3: Do wrong labels create tension, and does explanation type make it better or worse?

- **H3:** tension and frustration rise with wrong labels noticed per block.
- **H3a:** B and D lower tension even on wrong labels.
- **H3b (exploratory):** C raises tension when a ✗ row is visible.

Trust, fairness, time:

- **H4:** agreement and Jian trust are higher in B, C, D than in A.
- **H5:** fairness is rated highest in C and D, lowest in A.
- **H6:** C and D take longest; fact-sheet opens and row expands predict detection.

3. The Trial Screen

All conditions share one frame; only the explanation area changes, shown for p15 (a true post labeled misleading).

```
┌──────────────────────────────────────────┐
│ Part 3 of 4 · Post 2 of 6                │
├──────────────────────────────────────────┤
│ ┌─ POST (PostCard) ──────────────────┐   │
│ │ Riverton Neighbors ✔               │   │
│ │ @RivertonNeighbors · 6h            │   │
│ │                                    │   │
│ │ Good news: Riverton tap water met  │   │
│ │ all 12 safety limits in March.     │   │
│ │                                    │   │
│ │ ♡ 187   ↻ 52                       │   │
│ └────────────────────────────────────┘   │
│ ┌─ LabelPanel ───────────────────────┐   │
│ │ AI label: ⚠ Likely misleading      │   │
│ └────────────────────────────────────┘   │
│ ┌─ EXPLANATION AREA (fixed height) ──┐   │
│ │   varies by condition, see §5      │   │
│ └────────────────────────────────────┘   │
│                          [ Fact sheet ▸ ]│
│ [Agree with label] [Disagree with label] │
└──────────────────────────────────────────┘
```

A: No explanation

```
┌─ EXPLANATION AREA ─────────────────────┐
│                                        │
│            (empty: label only)         │
│                                        │
└────────────────────────────────────────┘
```

B: Static reason

```
┌─ EXPLANATION AREA ─────────────────────┐
│ Flagged: claims about water safety are │
│ often exaggerated.                     │
└────────────────────────────────────────┘
```

C: Interactive "Why?" (Source expanded)

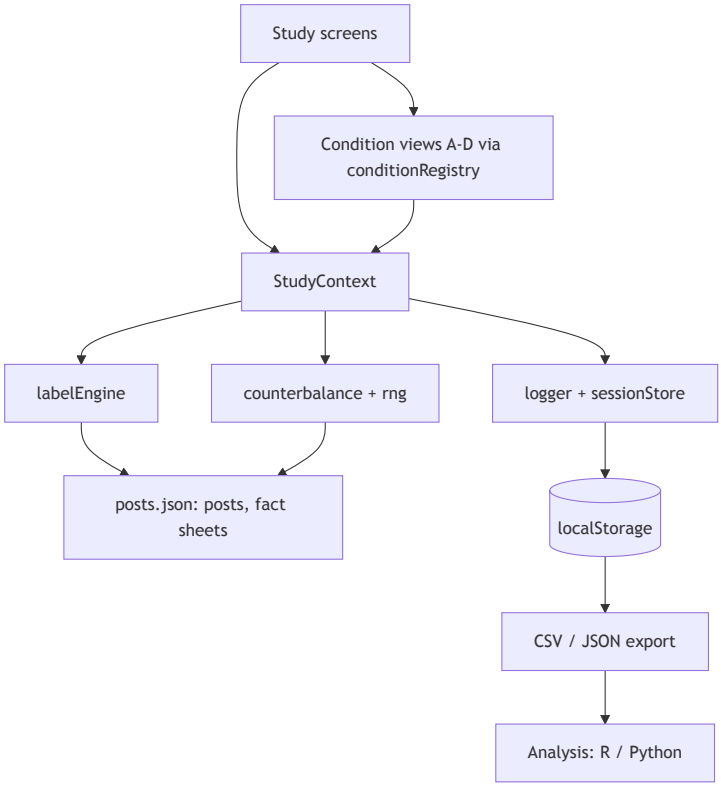
```
┌─ EXPLANATION AREA ─────────────────────┐
│ [ Why? ▴ ]                             │
│ ▾ Source                               │
│   @RivertonNeighbors · verified      ✓ │
│     Community pages can be verified    │
│     without being experts on water     │
│     testing.                           │
│ ▸ Claim vs record                      │
│   Post: all 12 limits met in March     │
│   Record: all 12 tests within limits ✓ │
│ ▸ Wording     Neutral                ✓ │
└────────────────────────────────────────┘
```

D: Counterfactual

```
┌─ EXPLANATION AREA ─────────────────────┐
│ This would be labeled accurate if it   │
│ cited the March test report.           │
└────────────────────────────────────────┘
```

4. Architecture

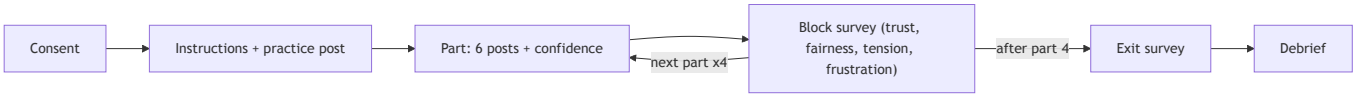
The app is a static React 18 + TypeScript + Vite SPA run in a moderated lab. A scripted label engine and pid-seeded counterbalancer feed one study state, saved to localStorage and exported as CSV/JSON.



- No backend, no database and no LLM: every label and explanation is hard-coded in `posts.json` (Wizard-of-Oz).
- Shared `PostCard`, `LabelPanel` and `FactSheetPanel`, with an identical fact sheet in every condition, so only the explanation varies.
- `conditionRegistry` maps A–D to four views, so `TrialScreen` never branches on condition.

5. Participant Flow

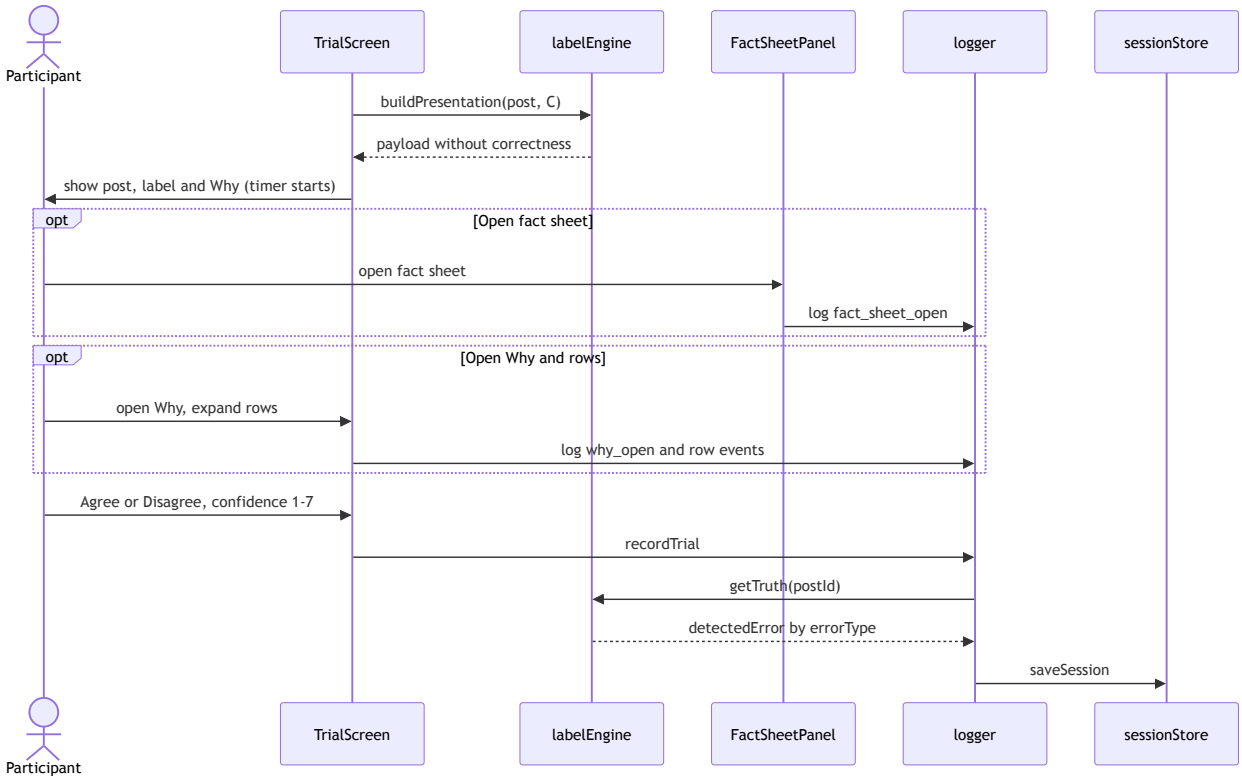
Each participant goes from consent through four parts of six posts, a short survey after each part, an exit survey and a debrief.



Condition order follows a 4×4 Williams square (`ABDC`, `BCAD`, `CDBA`, `DACB`) crossed with 4 post-set rotations, so full balance takes a **16-participant cycle**. The target is N = 32, with a minimum of 16. Within each part the six posts are shuffled, and the two wrong labels are never both in positions 1–2. After each part the block survey runs in a fixed order: 12 Jian trust items, 2 fairness items, tension (1–7), then the NASA-TLX frustration slider. A confidence rating is required after every post. A session takes about 30–35 minutes.

6. Key Logic: One Trial

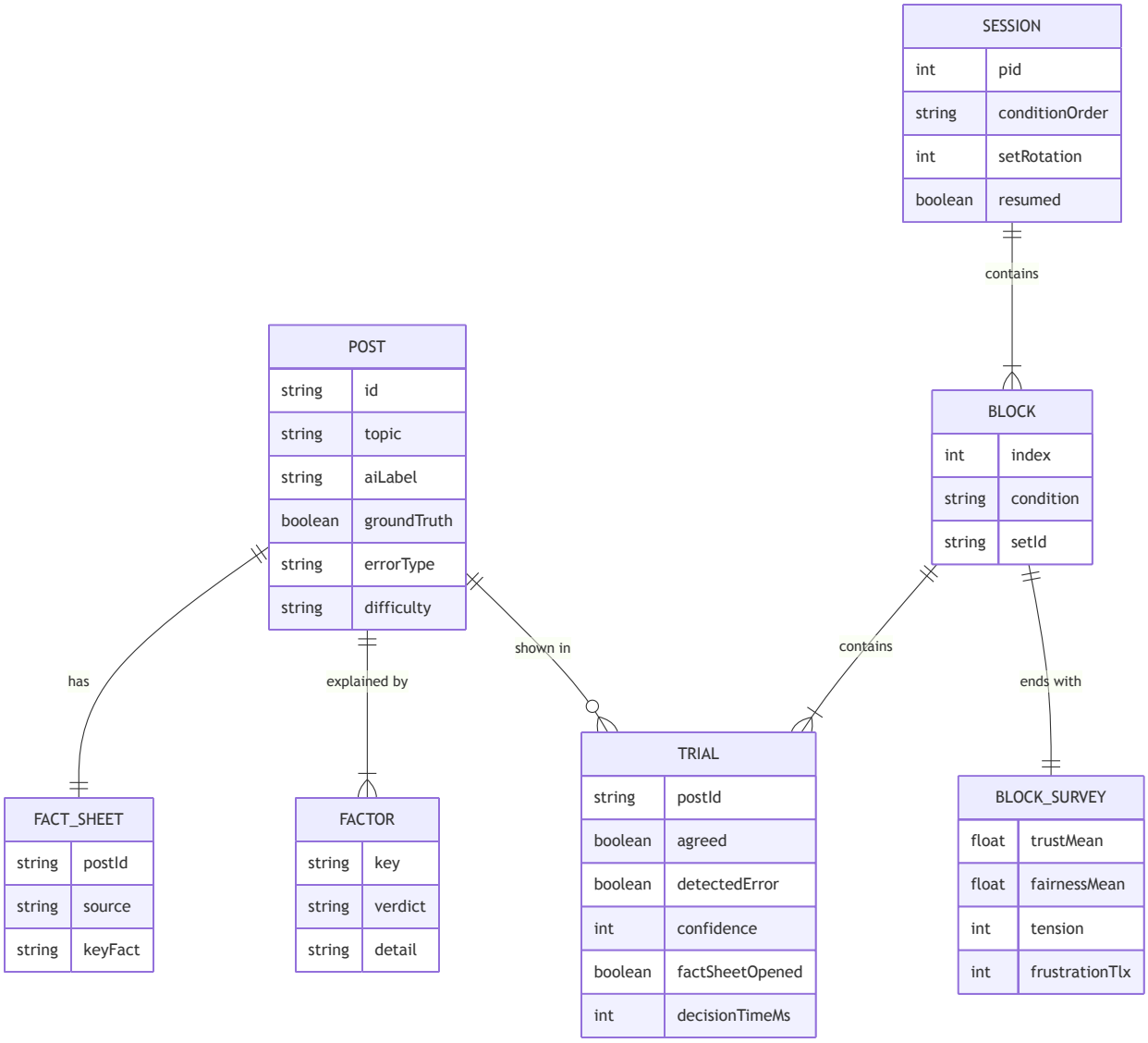
The label engine sends the view an allow-listed payload, so correctness never reaches the screen, and every label appears after the same ~800 ms "AI is checking…" delay. The fact sheet is identical across conditions and its use is logged, the timer runs from label shown to Agree/Disagree, and the session is saved after every trial.



- `detectedError = !agreed` on wrong labels (`null` on correct ones), analysed by `errorType`: `false_alarm` vs `miss` (H1a).
- Fact-sheet opens and time open are logged (H6).
- `timeToFirstExpandMs` records the time to the first "Why?" open (Condition C).
- A refresh resumes the session where it stopped, with `resumed = true`.

7. Data Model

There is no database: a read-only `posts.json` holds the posts with their fact sheets, and the session log lives in localStorage. The researcher exports `trials.csv`, `surveys.csv`, `events.csv` and `sessions.json`.

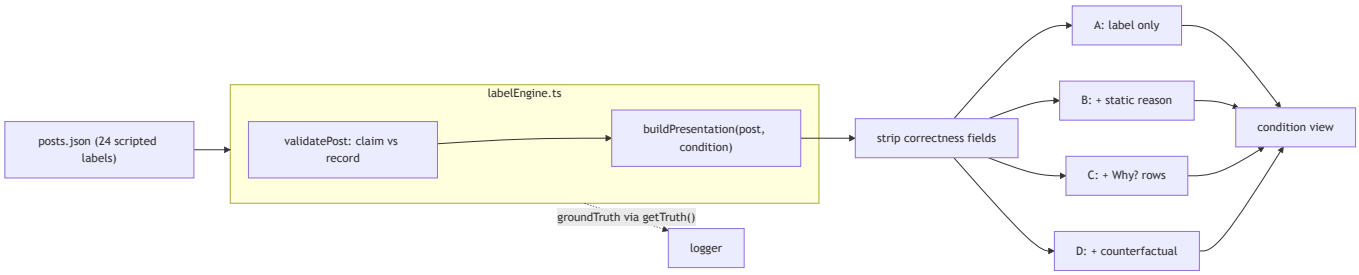


The 24-post bank

id	set	topic	post claim	fact sheet	truth	AI label	correct?	error type	difficulty
p01	S1	env	Recycling rate rose to 41% in 2025	2025 recycling rate: 41%	TRUE	accurate	Yes	—	—
p02	S1	svc	Library will close every Monday from May	Library closes Sundays only, open Mondays	FALSE	misleading	Yes	—	—
p03	S1	evt	Riverside Fair runs June 6–8 at Mill Park	Fair: June 6–8, Mill Park	TRUE	misleading	**No**	false_alarm	clear
p04	S1	env	Lake Alder hit a record low this spring	Lowest since 2019, record low was 2012	FALSE	accurate	**No**	miss	subtle
p05	S1	svc	Bus Route 9 adds Saturday service from March 1	Route 9 Saturday service starts March 1	TRUE	accurate	Yes	—	—
p06	S1	evt	Riverton marathon is cancelled this year	Marathon scheduled Oct 12	FALSE	misleading	Yes	—	—
p07	S2	env	City planted 1,200 street trees in 2025	1,200 trees planted in 2025	TRUE	accurate	Yes	—	—
p08	S2	svc	Trash pickup moves to every other week in April	Pickup stays weekly	FALSE	misleading	Yes	—	—
p09	S2	evt	Farmers market opens May 3 at Depot Square	Market opens May 3, Depot Square	TRUE	misleading	**No**	false_alarm	subtle
p10	S2	svc	Downtown parking now costs $4 an hour	Parking remains $2 per hour	FALSE	accurate	**No**	miss	clear
p11	S2	evt	Fireworks start 9:30 pm July 4 at the riverfront	July 4 fireworks 9:30 pm, riverfront	TRUE	accurate	Yes	—	—
p12	S2	env	Air-quality alerts on 30 days last summer	Alerts issued on 4 days	FALSE	misleading	Yes	—	—
p13	S3	svc	Library card renewals are now free online	Online renewal is free	TRUE	accurate	Yes	—	—
p14	S3	env	Alder River closed to swimming all summer	Open, 2 advisory days only	FALSE	misleading	Yes	—	—
p15	S3	env	Tap water met all 12 safety limits in March	March: all 12 tests within limits	TRUE	misleading	**No**	false_alarm	subtle
p16	S3	evt	Mill Park concert moved to August 20	Concert remains August 13	FALSE	accurate	**No**	miss	clear
p17	S3	svc	New crosswalk signals installed on Elm St	Elm St signals installed	TRUE	accurate	Yes	—	—
p18	S3	evt	Parade closes Main St for three days	Main St closed 1 day	FALSE	misleading	Yes	—	—
p19	S4	evt	Half-marathon raised $18,000 for the food bank	$18,000 raised	TRUE	accurate	Yes	—	—
p20	S4	svc	Property tax bills due two weeks earlier this year	Due date unchanged, May 15	FALSE	misleading	Yes	—	—
p21	S4	svc	Pool opening delayed to June 14 for repairs	Pool opens June 14 after repairs	TRUE	misleading	**No**	false_alarm	clear
p22	S4	env	City banned gas leaf blowers from this fall	Ban proposed, vote in November	FALSE	accurate	**No**	miss	subtle
p23	S4	env	Compost drop-off opens Saturdays 8–noon	Saturdays 8 am–12 pm	TRUE	accurate	Yes	—	—
p24	S4	evt	Holiday market tickets cost $25 this year	Entry is free	FALSE	misleading	Yes	—	—

8. The Wizard-of-Oz Labeller

No ML model or LLM runs at any point: every AI label is scripted in `src/data/posts.json`, so the team controls exactly which labels are wrong, how, and how they are justified. When the bank loads, `validatePost` checks each post's ground truth against its fact sheet's key fact, and the app refuses to start if any check fails. `buildPresentation` then strips the correctness fields before the view sees anything, and no AI confidence score is ever shown.



- **B, static reason:** one fixed sentence that argues for the label in the same confident tone whether it is right or wrong. For p15: "Flagged: claims about water safety are often exaggerated."
- **C, interactive "Why?":** three rows (Source, Claim vs record, Wording). The "Claim vs record" row shows the *true* comparison, so a wrong label can be checked from the row (team to confirm).
- **D, counterfactual:** one "would be … if …" sentence. For p15: "This would be labeled accurate if it cited the March test report."
- **Errors:** 4 false alarms (a true post labeled "Likely misleading") against 4 misses (a false post labeled "Likely accurate"), each with 2 clear and 2 subtle cases. Because error type and difficulty are fully crossed, the H1a headline comparison is not confounded with difficulty.

9. Analysis Plan

All analysis runs outside the app, in R (`lme4`, `emmeans`) or Python. With N = 32 the wrong-label tests have modest power, so they are reported as estimates with 95% CIs. H1b, H3b and D in H2 are exploratory.

Hypothesis	Measure	Test
H1: detection differs by condition	`detectedError` on the 8 wrong-label trials per pid	Mixed logistic `condition * errorType + (1|pid) + (1|postId)`, LRT on condition
H1a (headline): misses caught less than false alarms	as H1	Main effect of `errorType`, as an OR with 95% CI; `difficulty` as covariate
H1b: error-type gap smallest in C	P(detect | false alarm) − P(detect | miss)	`condition × errorType` interaction; C vs mean of A, B, D (`emmeans`)
H2: C highest, B lowest, D exploratory	as H1	Planned contrasts C > A, B, D and B < A (Holm); D vs A two-sided
H3: tension and frustration rise with errors noticed	`tension` (1–7), `frustrationTlx` (0–100) per block	Two LMMs `~ condition + nWrongDetected + (1|pid)`; slope > 0
H3a: B and D lower tension on wrong labels	as H3	Contrasts B < A and D < A, read alongside H2
H3b (exploratory): C raises tension when a ✗ row is visible	as H3, plus `factorsExpanded`	C > A on tension; C blocks with vs without a viewed ✗ row
H4: agreement and trust higher in B, C, D than A	`agreed`; mean Jian trust (items 1–5 reversed)	RM-ANOVA with Holm contrasts vs A; mixed logistic `agreed ~ condition * isAiCorrect`
H5: fairness highest in C and D, lowest in A	Mean of 2 fairness items	RM-ANOVA; contrasts C, D > A and C, D > B
H6: C and D slowest; clicks predict detection	log `decisionTimeMs`; `factSheetOpened`, expansions	LMM on time (C, D > A, B); mixed logistic `detectedError ~ factSheetOpened + condition`

10. Plan, Team & Next Steps

Phase	Weeks	Output
0 Setup	0	A repo that builds, lints and passes one test
1 Data & engine	1	`posts.json`, `labelEngine`, `counterbalance`, with test suites passing
2 Shared UI	1–2	Shared components in a gallery route, with the same layout across conditions
3 Condition views	2–4	Views A–D render any post
4 Study flow + surveys	3–5	The full flow from consent to debrief
5 Logging, export, admin	4–5	Exports that load in R and pandas
6 Pilot + fixes	5–6	Pilot log and a frozen build
7 Data collection	6–8	At least 16 sessions (target 32)
8 Analysis + submission	7–8	Report, video and `.zip`

Member	Area
Deepthi Reddy Chelladi	Condition A + shared trial components
Siwani Sah	Condition B
Bibek Gupta	Condition C
Afzal Hossain Sohag	Condition D + debrief content
Md Shamimul Islam	Post bank + engine + counterbalancing + logger
Mohamed Sylla	Study flow + surveys + storage/export/admin

Open decisions

- Prof. Hamam's approval of the topic change (new domain, condition D, and the tension and frustration measures).
- Ethics approval for the deception and the stress measures.
- The full literature search this week (Google Scholar, ACM DL), with Bibek Gupta as the proposed owner and Siwani Sah as second reader.
- Made-up posts only: does the team agree?
- Should C's "Claim vs record" row show the true comparison?
- The optional consequence banner and anxiety scale stay off by default unless the team adds them.

Target venues: in the near term, CHI Late-Breaking Work, IUI posters or short papers, or CSCW posters; later, a CHI or IUI full paper with a larger replication. Check the current calls for papers before planning around any deadline.

Deliverables: a source `.zip`, a demo video, and a report with screenshots of the code, the running app, and group members using it.